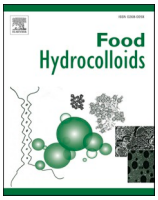
Food Hydrocolloids 158 (2025) 110552



Contents lists available at ScienceDirect

Food Hydrocolloids

journal homepage: www.elsevier.com/locate/foodhyd





The dissolution of galactomannans in *type II/III* deep eutectic solvents: Effect of polysaccharide structure and interaction mechanism

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ARTICLE INFO

Keywords:
Galactomannans
Deep eutectic solvents
Dissolution mechanism
Computational quantum chemistry

ABSTRACT

Deep eutectic solvents (DESs) have garnered increasing attention as sustainable solvents for natural polysaccharides. Galactomannans (especially guar and locust bean gums) are abundant natural compounds used in food, medicine, agriculture and daily chemical products, whereas their high molecular weight and randomly distributed non-linear structures remain a huge challenge in solubilization and high value-added utilization. In this study, the dissolution of galactomannans in *type II/III* DESs was comprehensively investigated using density functional theory (DFT) calculation, experimental results, and molecular dynamics simulation (MD). The results indicated that immense anionic and neutral H-bonds supported by *type II* DESs (consisting of a quaternary ammonium salt and a metal chloride hydrate) could break and rebuild galactomannan H-bond networks, thereby exhibiting remarkable physical solubilization. A large number of active acid sites in *type III* DESs (consisting of a quaternary ammonium salt and a hydrogen bond donor) greatly improved solubility and cleaved the glycosidic bond, enabling the recovery of polysaccharides with a target molecular weight. Furthermore, significant correlations between solubility and the structure of galactomannans had been established, which meant that high molecular weight and rich galactose side-chains produced repulsive interactions with DESs, directly hindering dissolution. Overall, this work not only offered a framework for enhancing the dissolution of polysaccharides, but also proposed a promising strategy for development of task specific DESs promoting green chemistry and the use of sustainable bioresources.

1. Introduction

The application of natural polysaccharides in food, medicine, materials and fuels have become an attractive option over the past few decades to mitigate increasing demands for resources in modern society and the promotion of environmental protection (Schlemmer, Selinger, Hobisch, & Spirk, 2021). Due to their biocompatible and biodegradable properties, polysaccharides have been utilized as building blocks to yield polysaccharide-based hydrogels through physical and/or chemical crosslinking of polysaccharides via a variety of monomers or ions. These polysaccharide-derived hydrogels exhibit peculiar physicochemical properties and excellent mechanical properties due to their unique structures and abundant functional groups. (Manzoor et al., 2022). Additionally, cellulose, starch and gel polysaccharides (e.g. galactomannans) as structural polysaccharides, are commonly used in biomedical fields including bone tissue engineering, cosmetic, food, antibacterial agents etc (Gopinath et al., 2022; Prajapati et al., 2013). Attentions have been paid by galactomannans which exhibit bioactivities such as reducing calorie intakes, controlling blood glucose, and reducing the risks of heart diseases and colon cancer (Li & Nie, 2016; Singh et al., 2021). Galactomannans possess a linear β-1,4-linked D-mannose backbone to which D-galactose residues are linked by α-1,6 glycosidic linkages (Moreira & Filho, 2008). In galactomannan family, the chemical structure for each member varies depending on the ratio of mannose to galactose (M/G). For example, the M/G ratio is approximately 4:1 for locust bean gum (LBG, from *Ceratonia siliqua*) and 2:1 for guar gum (GG, from *Cyamopsis tetragonolobus*) (Liu et al., 2020; Niknam, Mousavi, & Kiani, 2022). Due to the degree of galactose substitution, the

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https://doi.org/10.1016/j.foodhyd.2024.110552
Received 9 April 2024; Received in revised form 13 August 2024; Accepted 19 August 2024
Available online 23 August 2024
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fine structure of galactomannans is composed of "smooth" zones (lowly substituted) and "hairy" zones (highly substituted), which leads to a dense intra/inter hydrogen bonds networks (Grisel, Aguni, Renou, & Malhiac, 2015). As a result, it is difficult to dissolve in H_2O (highly swollen) and traditional organic solvents (dimethyl sulfoxide, formamide, etc.) that suffers from volatility and flammability, which makes the task specific functionalization very difficult (Depoorter et al., 2019). Thus, a novel, efficient and green solvent system for the functional proliferation of galactomannan is in urgent demand.

Nowadays, deep eutectic solvents (DESs) with low volatility, high designability and low toxicity are considered to have striking application potential in the processing of natural polysaccharides (Zdanowicz, Wilpiszewska, & Spychaj, 2018). A large number of studies have shown that DESs can achieve efficient solubilization of polysaccharides compared with traditional solvents due to their green characteristics and adjustable physical and chemical properties (Liu et al., 2023). The application of DESs in polysaccharide extraction has great advantages which increases the extraction rate of polysaccharides compared with traditional solvents (Table S1), as reported in extraction of *Lycium barbarum* (1.74-fold higher) and *Schisandra chinensis* polysaccharides (1.39-fold higher) (Gu, Lin, & Zhao, 2023; Yan et al., 2023). This can be explained by the fact that DESs featured by multiple hydrogen bond sites, which provide the abilities to occur hydrogen bonding interaction between DESs and polysaccharides, thereby achieving the purpose of efficient polysaccharides extraction (Pan et al., 2022; Zheng, Hu, Tu, & Wu, 2023). In addition, the task specific functionalization of natural polysaccharides by DESs is an important way to further its high-value utilization. Recent studies have shown that different types of surface chemical functionalization on the cellulose nanofibrils are obtained by using DESs to swell cellulose fibers and weaken of hydrogen bonding network, which confer to the final material distinct properties (Almeida, Maloney, & Gamelas, 2023). Meantime, our previous studies have also confirmed DES/DMSO binary mixed solvents boosting Se-functionalization by enhanced solubilization of heteropolysaccharide, which is attributed to the disruption of original hydrogen bond networks of polysaccharide, allowing the DESs within the system to interact more easily with polysaccharides (Sun et al., 2022). As is well-known, polysaccharides possess very high structural specificity, including molecular weight, monosaccharide composition, conformation, etc. However, it is important to note that most studies focus primarily on the application of DESs in the field of polysaccharide processing, with limited research conducted on the correlation of DES components with the structure of polysaccharides and the potential dissolution differences (Guo et al., 2021; Prasad & Sharma, 2019). In addition, some researchers have shown that acidic DESs can cause the degradation of polysaccharides, namely, the breaking of glycosidic bonds and the decrease of molecular weight (Yang et al., 2022). One such example is a study by Tian et al. (Tian et al., 2020), who used ChCl-oxalic acid (ChCl-OA) to treat cellulose. As a result, the acidity in ChCl-OA can provide H protons, thus breaking the glycosidic bond and finally obtaining several oligosaccharides linked by β-1,4-glucosidase bonds. The primary issue, therefore, is the relative lack of research on the structure of recovered polysaccharide after DESs treatment. Increased attention on the importance of this topic should attract a research model that systematically compares the dissolution relationship between DESs and recovered polysaccharide structure, to prove the physical or chemical dissolution of polysaccharides by DESs.

As discussed above, DESs have the advantage of designability. Based on this feature, an approach of expanding DESs function is the introduction of Lewis acid into the conventional *type II* DESs to prepare *type III* DESs, which is considered to have great application potential in the field of biomass processing. Using ChCl/glycerol (1:2) as an example, Xia and co-workers show that this system is unable to cleave ether bond linkages in lignin-carbohydrate complexes due to the absence of active protons and acidic sites. Accordingly, based on an acidic multisite coordination theory, the *type III* DESs are designed. The competition of anionic H-bonds and unidentate aluminum ligands is synchronized to form supramolecular complexes, allowing the multisite bridging ligands to cleave both the H-bonds and ether bonds in lignin-carbohydrate complexes, and the purity of extracted lignin is as high as 94 % (Xia et al., 2018). Additionally, high viscosity is one of the main obstacles to the further application of DESs. The addition of aprotic solvent as the third component to *type II* DESs can effectively reduce the viscosity of conventional DESs for the further processing and utilization of polysaccharides. For example, the addition of DMSO as a cosolvent to traditional *type II* DESs can reduce the viscosity and increase the reactivity, which greatly improves the effective conversion of sugar bagasse into platform chemicals (Amesho et al., 2022). Almost simultaneously, it is reported that the conversion rate of hexose in corn stover was significantly increased from 10.95 % to 41.40 % by using DMSO to reduce the viscosity of ChCl:oxalic acid (Lu et al., 2022). However, given the immense number of HBAs and HBDs, finding "optimal" eutectic mixtures for a given application will currently exhaustive trial-and-error through millions of possible binary combinations (Hansen et al., 2020). This is one of the reasons why the current *type III* DESs are mostly limited to the treatment of cellulose, and the application of other polysaccharides is less. Moreover, on the one hand, the *type III* DESs by introducing Lewis acid will form strong anions ($OH\cdots Cl^-$) and neutral hydrogen bonds ($OH\cdots O$), which will break the glycosidic bond of polysaccharides (Ji, Yu, Wu, et al., 2021; Zhou, Fakayode, Yagoub, Ji, & Zhou, 2022); on the other hand, it responds to the molecular weight, branching degree and crystallinity of complex polysaccharides, resulting in different dissolution results. While *type III* DESs have demonstrated excellent properties in extracting cellulose from biomass, research on *type III* DESs is still in its early stages (Bai, Zhang, Wang, Zheng, & Yao, 2022; Zhang, Wu, Wang, Li, & Fang, 2024). Particularly, reports on the potential solubilization ability of *type III* DES in polysaccharides treatment, are rare. Moreover, the influence of *type III* DESs composition and proportion on the structure of the recovered polysaccharide are not well elucidated, which hinder the development of DESs and their utilization in polysaccharides dissolution and functionalization.

As such, a primary objective of current research is to comparatively evaluate the dissolution relationship between polysaccharide structures and DES components, and to adapt DESs for specific applications. In this work, two typical galactomannans (LBG and GG), with same linear β-1,4-linked D-mannose backbone but different M/G and molecular weight, were employed to study the effect of molecular weight and distribution pattern of Gal substituents on DESs dissolution. First, density functional theory (DFT) calculation was performed to predict the dissolution effects of GG and LBG fragment models in several traditional *type II* DESs. After preparing and characterizing a series of *type II* DESs, the optimal *type II* DES was screened from viscosity, polarity and solubility to LBG and GG. Subsequently, the dissolution mechanism of *type II* DESs on galactomannans was elucidated by DFT, molecular dynamics simulation (MD) and structural characterization of recovered polysaccharides. Taking advantage of DESs designability, *type III* DESs with $AlCl_3{\cdot}6H_2O$ as the third component were employed to achieve efficient dissolution of galactomannans. The effect of *type III* DESs with Lewis acid on the structure of the galactomannans during the solubilization was further evaluated by structural characterization of recovered galactomannans. This study aimed to explore the dissolution relationship between polysaccharides structure and DES components, to provide reference for the realization of DESs task specific functionalization and to fully utilize polysaccharide resources.

2. Materials and methods

2.1. Materials

Choline chloride (ChCl), ethylene glycol (EG), urea, 1,4-butanediol (BDO), glycerin (GLY), potassium bromide (KBr), *N,N*-dimethyl-4-nitroaniline (DMNA) and 4-nitroaniline (NA) were purchased from

Jingchun Industry Co. Ltd. (Shanghai, China). In this study, all of the above reagents were of analytical grade.

Guar bean gum (GG) and locust bean gum (LBG) were purchased from Sigma-Aldrich (USA) with the weight average molecular mass (M_W) of 2.25 × 10^6 g/mol and 8.65 × 10^5 g/mol, and polydispersity (PD) of 1.098 and 1.245, respectively. The monosaccharide composition of GG and LBG were D-mannose and D-galactose (M/G ratio of 1.20:1 and 3.63:1, respectively).

2.2. Computational methods

Initial geometry optimizations of DESs and galactomannan model compounds were obtained using the B3LYP exchange-correlation function combined with the 6-311G (d, p) basis set (Pritchard, Altarawy, Didier, Gibson, & Windus, 2019; Stephens, Devlin, Chabalowski, & Frisch, 1994). To eliminate the dispersion effect, we added the key word of empirical dispersion = GD3BJ in the route section. Then, the initial structure of optimized DESs were performed at HF/6311G (d, p) level, and further reoptimized using the M06-2X exchange-correlation function in combination the 6–31++G (d, p) basis set based on the Gaussian 16 program package (Walker, Harvey, Sen, & Dessent, 2013). Given Gaussian results, the electrostatic surface potential (ESP) (Zhang & Lu, 2021), atom in molecules (AIM) (Bader, 1985), reduced density gradient (RDG) (Johnson et al., 2010) and frontline molecular orbitals (FMO) theory analyses were performed using the Multiwfn 3.8 programs (Lu & Chen, 2012). All the ESP, AIM and FMO maps were rendered by Visual Molecular Dynamics (VMD) 1.9.4 program. The data of RDG was visualized by VMD 1.9.4 and Gnuplot 5.4.3 software.

All molecular dynamics (MD) details were performed in Gromacs 2022.3, and all studied molecules were parameterized using the charm 36 force field (AitRaise et al., 2023; Xu, Kong, Peng, Wang, & Li, 2021). The (4M-1G)-C1E4 system consisted of 186 ChCl molecules, 744 EG molecules and one 4M-1G models molecular chain. The C1E4 components were added to a 6 nm × 6 nm × 6 nm box in a certain amount and proportion. Periodic boundary condition (PBC) was applied to the three directions of the box to ensure that the number of particles in the box was constant. Under certain temperature and pressure conditions, the steepest descent method was used to simulate the system in 50000 steps to minimize the energy of the system. Then the systems were equilibrated for 100 ps under NVT ensemble with temperature at 300 K. Following was a 100 ps simulation under NPT ensemble to equilibrate the solvents. Finally, a long-term MD simulation with a step size of 250 million steps and a duration of 500 ns was carried out at constant temperature and constant pressure.

2.3. Preparation of DESs

First, all relevant materials were dried before preparation. Then, mixing HBAs with corresponding HBDs according to the molar ratio provided in Table S3 to prepare *type II* DESs. The mixture in the sealed bottle was stirred with a magnetic stirrer and heated at 80 °C until clear homogeneous liquid was obtained (Ramezani, Ahmadi, & Yamini, 2022).

The preparation of *type III* DESs entailed weighing the corresponding components according to their composition and proportion, followed by stirring at 80 °C until a homogeneous and transparent liquid was formed.

2.4. Polarity determination of type II DESs

Polarity scale (*EN T*), π* (dipolarity/polarizability), β (hydrogen bond basicity) and α (hydrogen bond acidity) values were calculated according to modified *Kamlet-Taft* empirical polarity scales (Ranjkesh, Parast, Strzeżysz, Zakerhamidi, & Yoon, 2018).

2.5. Measurement of viscosity

The viscosity of the polysaccharide solution was measured using an Anton Paar Physica MCR 301 Rheometer (Anton Paar, GmbH, Austria) equipped with a cone-plate geometry (50 mm diameter). Samples were loaded onto a smooth Peltier plate which maintained constant experimental temperatures of 90 °C (±0.1 °C) for the studies presented herein. The plate spacing was 1 mm and temperature equilibration of the samples was achieved within 5 min prior to each measurement (Potier, Tea, Benyahia, Nicolai, & Renou, 2020). For each polysaccharide solution, steady shear and dynamic oscillatory shear experiments were carried out in the shear rate range of 0.1–1000 s^{-1} and frequency range of 0.1–100 rad/s, respectively. All rheological data were analyzed with rheoplus software (Rheoplus/32 V3.40).

2.6. Solubility of galactomannans in different solvents

The detailed measurement procedure for the solubility of galactomannans was based on the study by Cao et al. (Cao, Nian, Li, Wu, & Liu, 2019). Specifically, the dissolution process involved accurately weighing galactomannans (4 g ± 0.1 mg) into different solvents (60 mL ± 0.1 mL) and stirring the mixture magnetically at 80 °C for 5 h. The dissolved sample was centrifuged at 8000 rpm for 30 min, and the supernatant was collected. The supernatant was diluted 50 times with deionized water. Then, 1 mL of deionized water and 1 mL of 6% (w/v) phenol were added and mixed thoroughly. Next, 5 mL concentrated sulfuric acid was added to the mixture and fully mixed. After standing for 30 min, the absorption spectrum of the mixture at 490 nm was measured and calculated according to the following equation:

$$S = (A_{490nm} - b)/a \times 0.9 \times 10 \quad (1)$$

Where S represented the solubility of galactomannans (g/100 g), the absorbance of the mixture at 490 nm was expressed as A_{490nm}, the slope and intercept of the standard curve were expressed as a and b. The standard curve was obtained by dissolving 0.1, 0.2, 0.3, 0.4 and 0.5 g of glucose into 10 g of different DESs according to the above procedure. The determination process was performed in triplicate. The experimental data were analyzed using SPSS V.17.0 ($p < 0.05$).

2.7. Polarized light microscope

4 g GG and LBG were put into 60 mL different DESs, respectively, and stirred at 80 °C for 6–8 h. Dip a drop of GG solution onto a glass slide with a glass rod and observe it with a polarizing microscope (Axio scope A1, Carl Zeiss, Germany).

2.8. Fourier transform infrared (FT-IR) spectroscopy

The spectra were recorded at the wavenumber range of 4000–400 cm^{-1} with FT-IR spectrophotometer (iS10, Thermo Nicolet, USA). The number of scans was 16 and referenced against air. The resolution of FT-IR was 4 cm^{-1} (Beratto-Ramos, Agurto-Muñoz, Vargas-Montalba, & Castillo, 2020).

2.9. NMR spectroscopy

^1H and $^1H-^1H$ COSY NMR spectra were recorded at 25 °C by a Bruker Avance DPX-400 spectrometer (Bruker, Rheinstetten, Germany). Chemical shifts were reported in ppm using TMS as the internal reference. The NMR spectra of DESs and polysaccharides were determined in DMSO-d_6 and D_2O, respectively (Holding, King, & Kilpeläinen, 2018).

2.10. Monosaccharide composition analysis

The sample (10 mg) was hydrolyzed with trifluoroacetic acid under

nitrogen protection at 120 °C for 8 h. The residual trifluoroacetic acid was removed by rotary evaporation. The hydrolysis product was added with ammonium hydrochloride (10 mg) and pyridine (1 mL) and reacted at 90 °C for 30 min. After cooling to room temperature, 1 mL of acetic anhydride was added and reacted at 90 °C for 30 min. The final product was extracted with 1 mL of chloroform and filtered through a 0.22 μm filter. The ensuing analysis was conducted by gas chromatograph (GC7900, Shanghai Tianmei Scientific Instrument Co., Ltd.).

2.11. Molecular mass analysis

The molecular mass of the sample (1 mg/mL) was determined via size-exclusion chromatography combined with multi-angle laser light scattering (SEC-MALLS). The sample was filtered through a 0.45 μm cellulose filter and eluted by an Ultrahydrogel™ column (7.8 × 300 mm, Waters, USA) with ultrapure water as mobile phase. The injection volume and flow rate were 50 μL and 0.5 mL/min, respectively. Zimm method was used to calculate the weight average molecular mass (M_W) and data analysis was performed using Astra software (Wyatt Tech. Corp., USA).

3. Results and discussion

3.1. Dissolution of galactomannans in type II DESs

The effect of molecular weight and distribution pattern of Gal substituents dissolution on DESs dissolution have attracted great attention. The hydrogen bonding strength in galactomannans and traditional *type II* DESs was comparative studied by the atom in molecule (AIM) theory, thereby evaluating the potential solubility of GG and LBG (Fig. 1). The H-bonding strengths of traditional *type II* DESs and galactomannans were calculated at atomic level using 4M-2G and 4M-1G as sugar chain fragment models for GG and LBG. Typically, electron density (ρ) and its Laplacian ($^{\nabla 2}\rho$) could quantify the properties of hydrogen bonds which was shown in ranges from 0.02 to 0.15 au and 0.002 to 0.04 au for $^{\nabla 2}\rho$ and ρ, respectively (Lipkowski, Grabowski, Robinson, & Leszczynski, 2004). The ρ values of 4M-2G and 4M-1G models were 0.035 ~ 0.005 au and 0.037 ~ 0.004 au (Table S2), which were larger than those of ChCl-2EG (C1E2, 0.026 ~ 0.008 au) and ChCl-2Urea (C1U2, 0.023 ~ 0.017 au). Furthermore, the absolute values of hydrogen binding energy ($|E_{HB}|$) for 4M-2G and 4M-1G models, C1E2 and C1U2 were calculated (Fig. 1) and employed to reflect the types of hydrogen bond strength (Emamian, Lu, Kruse, & Emamian, 2019). According to the $|E_{HB}|$, it could be divided into strong ($|E_{HB}|$> 15.0 kcal/mol), medium (11 < $|E_{HB}|$ < 15 kcal/mol), weak to medium (2.5 kcal/mol <$|E_{HB}|$ < 14.0 kcal/mol) and weak ($|E_{HB}|$ < 2.5 kcal/mol) H-bond. $|E_{HB}|$ of C1E2 and C1U2 had weak to medium H-bonds, which were higher than 4M-2G and 4M-1G models. Based on the above results of AIM and $|E_{HB}|$, it was predicted that C1E2 and C1U2 could break the hydrogen bond network of LBG and GG, thus reflecting a certain solubility. These findings were consistent with the results of previous research. Those studies, utilizing AIM, predicted that the ionic liquids, 1-allyl-3-methylimidazolium chloride and 1-butyl-3-methylimidazolium chloride, possessed the ability to disrupt hydrogen bonds, thereby facilitating the dissolution of

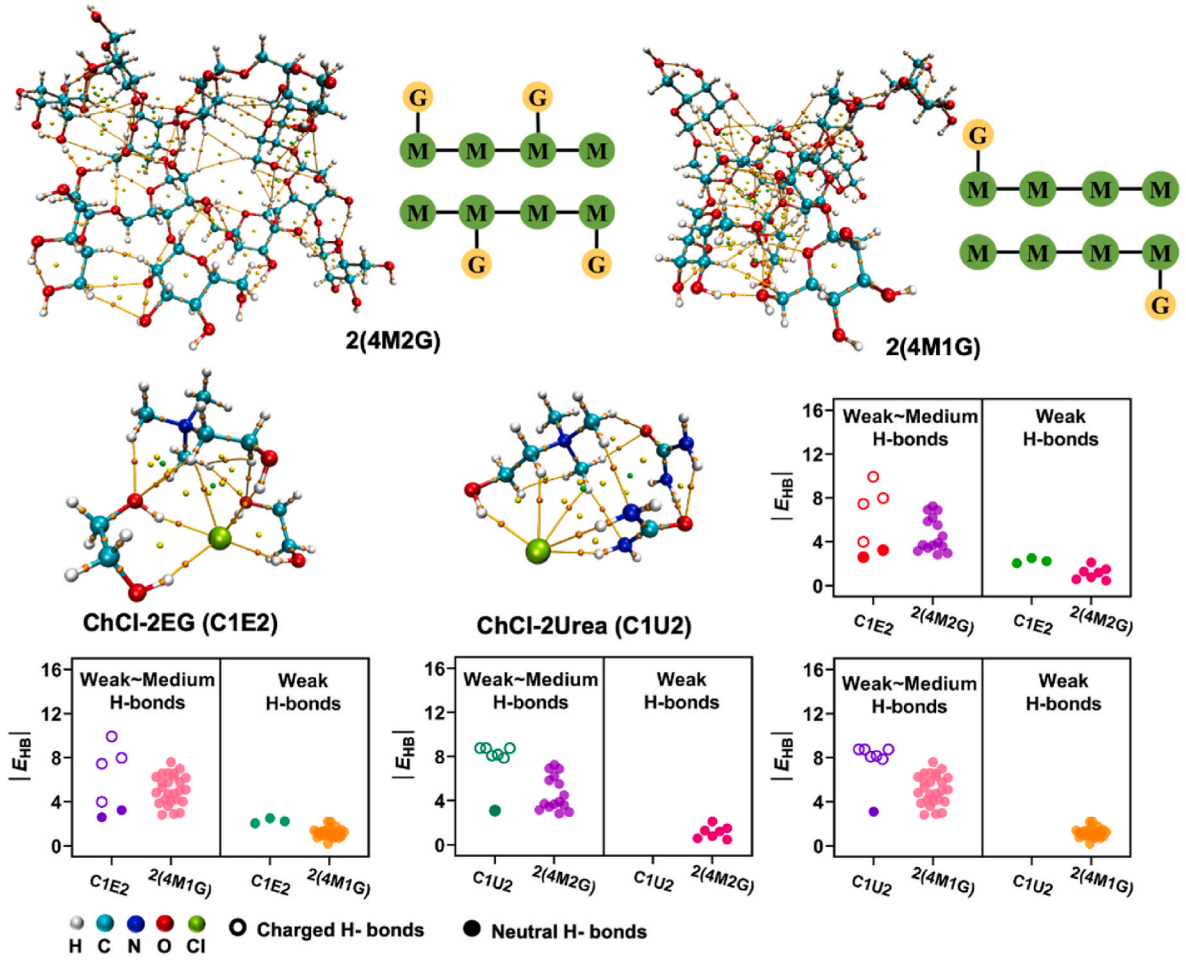


Fig. 1. AIM topological analysis plot of 4M-2G and 4M-1G models and traditional *type II* DESs, the absolute value of hydrogen binding energy ($|E_{HB}|$) for 4M-2G and 4M-1G models and traditional *type II* DESs. The data were calculated at the B3LYP-D3(BJ)/6-311G (d, p) level. All data of AIM were shown in Table S2.

galactomannan (Jia et al., 2024).

Next a series of *type II* DESs were synthesized with different HBDs to evaluate their solubility for galactomannans. FT-IR spectroscopy and ^{1}H NMR were used to explore the H-bond interaction of DESs. All details were listed in Fig. S1. In *type II* DESs, C1E4 was taken as an example to analyze the specific characterization results. It could be clearly observed from the FT-IR spectra that the hydroxyl groups on ChCl and EG had blue-shifted from 3255 and 3357–3374 cm^{-1}, respectively, indicating that the interaction between ChCl and EG had resulted in an increase in the vibration frequency of the chemical bond and the formation of new hydrogen bonds. Similar phenomena had been reported in research conducted by Feng et al. (Feng et al., 2022) and Zhu et al. (Zhu et al., 2016), where hydrogen bond interactions between HBD and HBA in DESs could lead to blue shifts in FT-IR spectra. This further confirmed the universality and significance of this phenomenon within DES systems. In addition, as shown ^{1}H NMR spectra, compared with ChCl, the hydrogen atom of H-1 shifted from 5.74 to 5.46 ppm. Meanwhile, the H-5 located on EG was found to be low field shifted from 4.52 to 4.56 ppm. These results illustrated the H-bond interaction between ChCl and EG.

Kamlet–Taft solvatochromic parameters were utilized as quantitative tool to measure the polarity of DESs (Fig. 2c and Fig. S3). These parameters effectively assessed the hydrogen-bond donating ability (acidity, α), hydrogen-bond accepting ability (basicity, β), and polarity/polarizability (π*) of C1E4. The comparison of C1E2 and ChCl–4EG (C1E4) showed that the α values of C1E4 was improved to 0.917, such high values indicated that it was easy to release H^{+}. The π* values exhibited a decrease to 1.077 suggesting that the solubility of C1E4 was higher compared to C1E2. Similar findings had also been reported by Zhang et al. (Zhang et al., 2023). Their results showed that lignin solubility had a closed positive correlation with the subtraction value of α and β of DESs. Among all the investigated DESs, L-alanine/formic acid (1:3) with the strong hydrogen-bond donating ability, weak hydrogen-bond accepting ability and small steric-hindrance effect showed the best lignin dissolving ability. Therefore, the H-bond accepting and dissolving capacity of the solvent increased after improving proportion of EG.

It was apparent that the solubilities of LBG and GG were greatly higher in *type II* DESs than in conventional solvent, the maximum solubility of GG was 0.28 g/100 mL and the minimum was 0.02 g/100 mL in *type II* DESs for instance. Surprisingly, the solubility of LBG was approximately 28 times higher compared to GG (Fig. 2a). Meanwhile, the viscosity followed the order: GG solutions > LBG solutions > *type II* DESs at the same temperature (Fig. 2c and Fig. S2). These results proved that galactomannans had different solubilization consequences in the same DESs, which was dependent on the molecular weight and degree of branching. Upon examination under a polarizing microscope (Fig. 2b), the solutions of LBG and GG in DESs exhibited complete dark field, signifying that LBG and GG were fully dissolved in the DESs, resulting in an optically transparent solution. In addition, the solution showed a uniform appearance with no visible particles or precipitates, further confirming the effective dissolution of LBG and GG in C1E2 and C1E4. This microscopic observation corresponds to the obtained solubility results, emphasizing the dissolution characteristics of galactomannan in

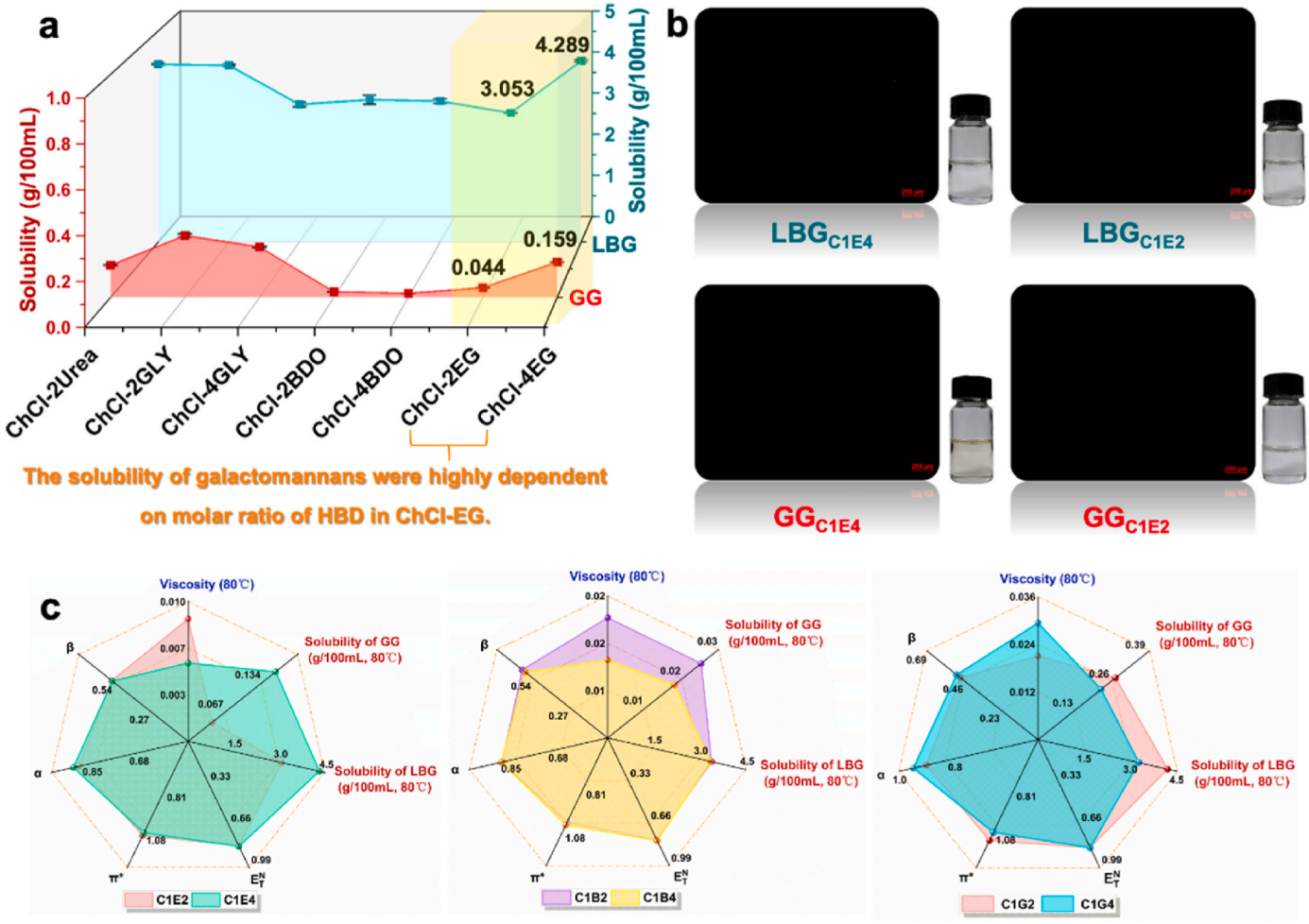


Fig. 2. (a) Solubility of LBG and GG in different *type II* DESs, (b) Polarizing microscope images of LBG and GG dissolved in ChCl-EG system, (c) The relationship between solubility, viscosity, and *Kamlet–Taft* parameters of *type II* DESs with the same HBD.

the selected DESs. Overall, EG of high proportion provided more hydrogen bond sites to break the original galactomannan hydrogen bond networks and rebuilt new hydrogen bonds, which suggested that C1E4 held remarkable solubility for galactomannans.

On the other hand, the characteristic absorption peaks of galactomannans, ChCl and various HBDs appeared in the FT-IR spectra of all galactomannan solutions in *type II* DESs (Fig. S4). As an example, two representative bands of urea corresponding to the N-H bending vibration and C-N stretching vibration, at 1669 and 1620 cm^{-1}, observed in ChCl–2Urea–GG and ChCl–2Urea–LBG. These results suggested that the basic structure of galactomannans, ChCl and various HBDs was not destroyed. Besides, compared with neat galactomannans, the hydroxyl peaks on galactomannan solutions shifted, indicating the occurrence of intermolecular H-bonding. These results illustrated the H-bonding interaction between *type II* DESs and galactomannans, and therefore promoted the solubilization. The study reported by Wei et al. also demonstrated that stable intermolecular hydrogen bonds could be formed between DES and cellulose (Wei, Long, Li, Gao, & Nie, 2024).

3.2. Investigation of H-bonding interactions of DES by DFT calculation

The different qualities of H-bonds in DESs lead to the difficulties associated with the quantification of H-bonding interactions. Here, DFT calculation was employed to gain an insight into the H-bonding interactions of DESs at an atomic level. The optimized structures of ChCl, EG and C1E4 were shown in Fig. 3a. The choline cation (Ch^+) might be considered as a tetrahedron formed by the methyl/methylene groups of a quaternary ammonium head and a CH_2OH chain extending from one of the vertices and partially blocking the corresponding face. Additionally, two hydroxyl groups in the EG endowed the C1E4 with numerous H-bond interacting capabilities. Thus, the main types of H-bonding in C1E4 identified as: cationic H-bond (Ch^+ and EG), anionic H-bond (Cl^- and EG), doubly ionic H-bond (ChCl ion pairs), and neutral H-bond (EG - EG).

AIM theory could quantify the H-bond interactions at bond critical points (BCPs) in accordance with ρ and $\nabla^2\rho$. The topological diagram of C1E4 exhibited by AIM investigation (Fig. 3a), and summarized the values of ρ and $\nabla^2\rho$ for all BPCs between ChCl and EG (Table S4). The positive values of $\nabla^2\rho$ were all within the H-bond and van der Waals force range. The results revealed that the $O_{41}-H_{42}\cdots Cl_{22}$ H-bond interaction with Cl^- was one of the strongest H-bonds ($\rho = 0.0255$ au and $\nabla^2\rho = 0.0619$). The EG molecule formed anionic and cationic H-bond with Cl^- and Ch^+ in ChCl, respectively. Hence, a large number of various strengths of hydrogen bonds formed in C1E4. Furthermore, noncovalent interactions in AIM were visualized by employing reduced density gradient analysis of the charges. The 2D/3D noncovalent interaction diagrams exhibited that the strong interaction H-bond and van der Waals interaction region of RDG was wider (Fig. 3c), which illustrated that the free sites of substantial Cl^- were occupied by the H-bonds. This finding was supported by the research of Gautam et al. that the chlorine ion in choline chloride was found to form a strong hydrogen bond with acetic acid and formic acid in their DES formed with choline chloride (Gautam, Kumar, & Lynam, 2020). Above studies had shown that C1E4 with a high proportion of EG theoretically provided more hydrogen bond sites to solubilize LBG and GG.

In addition, the electrostatic surface potential (ESP) was adopted to map electrostatic potential plots of the van der Waals surface as well as the local minima/maxima points of DESs, where the red color indicated the positive potential surface, the blue represented the negative ones and white areas meant electrical neutrality (Fig. 4). ESP analysis showed that the electrostatic potential distribution of C1E4 was not uniform, and the negative electrostatic potential region accounted for a relatively high proportion. The minimum value was −42.75 kcal/mol, and the maximum positive point was 40.04 kcal/mol. With the interactions of C1E4 and galactomannans, the positive and negative electrostatic potential played an important role in the formation of hydrogen bonds. Additionally, the numerous -OH of EG increased proportion of negative electrostatic potential for C1E4 compared with the C1E2 (Sun et al., 2022), which might contribute to extensive hydrogen bond interaction between C1E4 and galactomannans, thus improving the dissolution of LBG and GG. On the other hand, the frontier molecular orbital further revealed that the highest occupied molecular orbitals (HOMO) electron cloud of C1E4 was localized on near the methyl of ChCl, and the lowest unoccupied molecular orbitals (LUMO) electron cloud was mostly

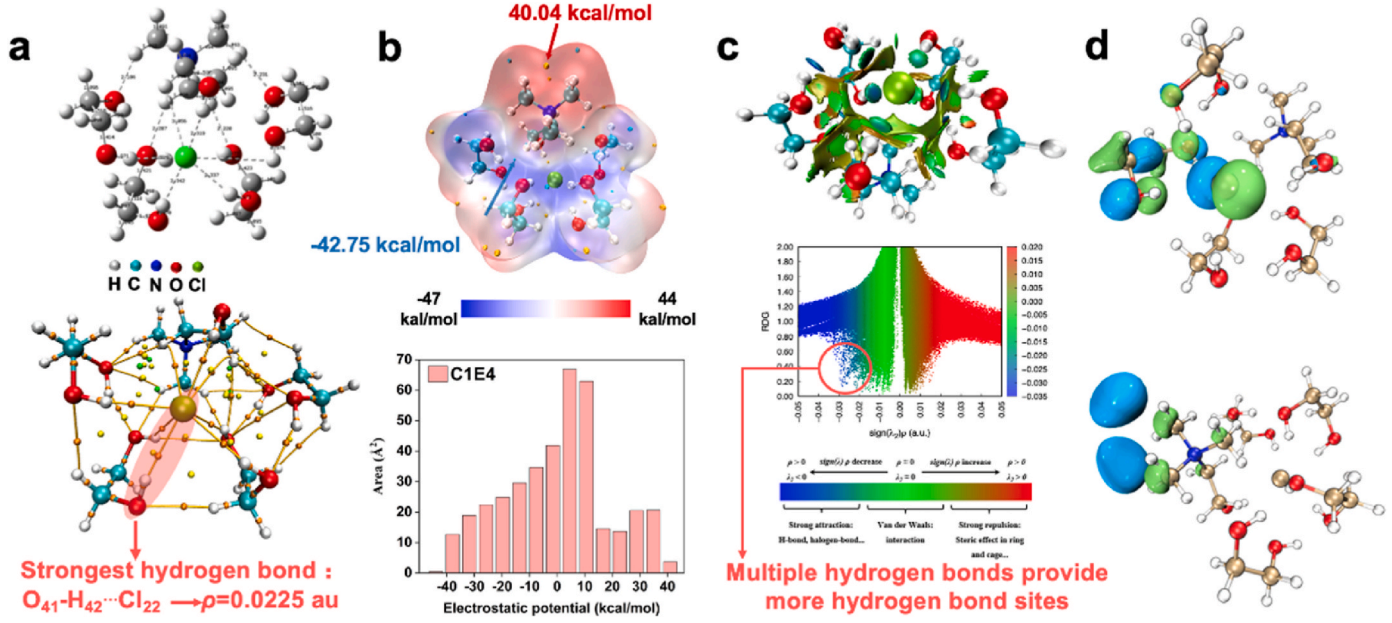


Fig. 3. (a) Optimized geometry and AIM topological analysis plot of C1E4, (b) Molecular electrostatic potential values mapped on vdW surface and the percentage of area within each ESP range. The minimum and maximum values of the significant surface localization of ESP were denoted by cyan and yellow spheres, (c) 2D/3D noncovalent interaction diagrams of C1E4, the large, negative values of sign(λ^2) ρ indicated hydrogen bond (blue), the large, positive values of sign(λ^2) ρ represented spatial repulsive (red), and the values close to zero were associated with vdW interactions (green), (d) HOMO and LOMO orbitals of C1E4. The data were calculated at the M062X/6–31++G (d, p) level.

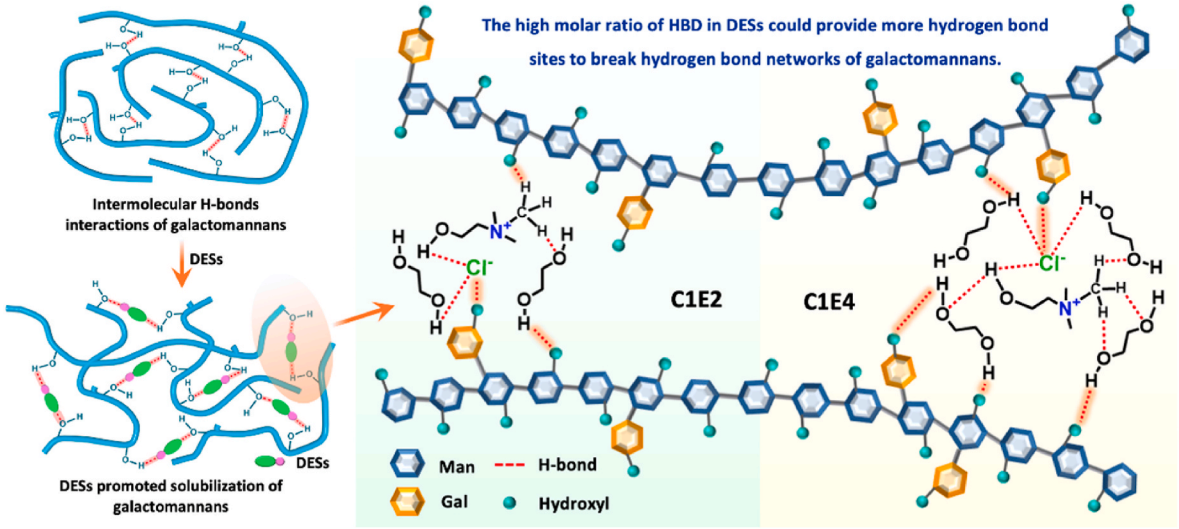


Fig. 4. The dissolution mechanism of galactomannans in C1E4 and C1E2.

localized on the carbon and hydrogen atoms of Ch^+ (Fig. 3d). Hence, it was speculated that the LUMO and HOMO regions in C1E4 might interact with the nucleophilic and electrophilic sites in LBG and GG, respectively. According to the above theoretical calculations, the extensive hydrogen bonding sites in the C1E4 system provided the ability to interact with LBG and GG. Similarly, Li et al. used DFT to explore the mechanism of ternary DES in the extraction of *Ganoderma lucidum* polysaccharides, and they revealed the significant contribution of multiple hydrogen bond sites of DESs to the extraction efficiency (Li, Shi, Chen, & Liu, 2024).

3.3. Dissolution mechanism of galactomannans in type II DESs

Through the analysis of FT-IR and ^{13}C NMR, chain characteristics of recovered galactomannans were explored. The FT-IR spectra of the original and the recovered galactomannans were shown in Fig. 5a. The O–H stretching vibration around 3420 cm^{-1} and the bending vibration of C–O–C and C–H region around 1030 and 1417 cm^{-1} observed in recovered galactomannans, which were often considered as typical saccharide band regions. It could be seen that no new peaks were observed in the recovered samples, which indicated that the basic skeleton of recovered galactomannans were not destroyed during the dissolution processes. Additionally, hydroxyl stretching vibration of recovered LBG blue-shifted when compared with original LBG (3421 cm^{-1}), revealing the breaking of native polysaccharide hydrogen bonds to some extent. The absorption band at 1431 cm^{-1} for C–H bending vibration in original LBG appeared a red shift in recovered LBG, suggesting the destruction of an intra-molecular hydrogen bond involving hydroxyl. For GG, the absorption band of hydroxyl group and C–H bending vibration at 3421 and 1431 cm^{-1} showed a small red shift in recovered GG due to the low solubility in *type II* DESs. The above consequences verified that the dissolution of galactomannans was attributed to hydrogen bonding interactions between *type II* DESs and galactomannans to break the original galactomannans hydrogen bond networks and rebuild new hydrogen bonds. Moreover, the structures of recovered LBG_{C1E2}, LBG_{C1E4}, GG_{C1E2}, and GG_{C1E4} were analyzed by ^{13}C NMR (Fig. 5b), and the carbon signals of the recovered polysaccharides of the LBG and GG series were assigned. Taking recovered LBG as an

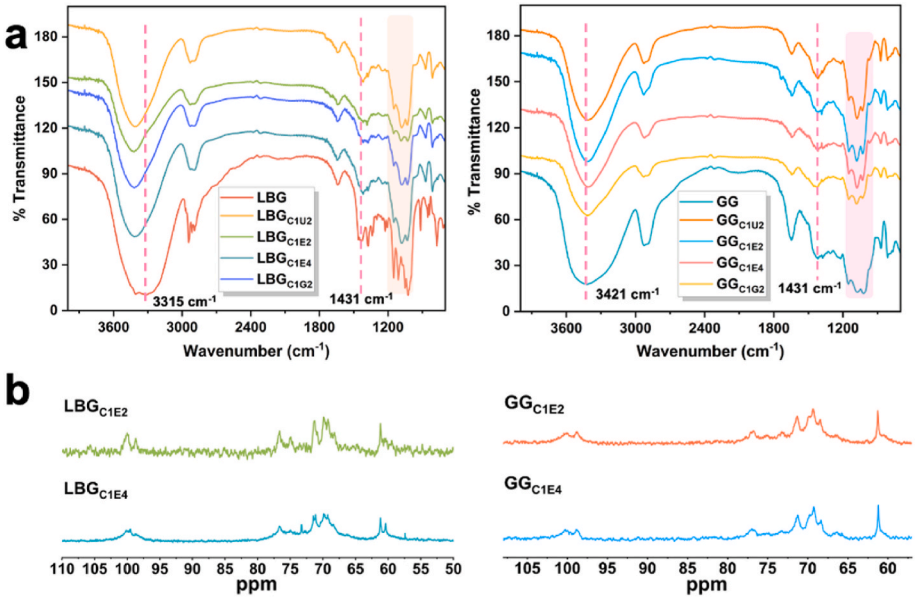


Fig. 5. FT-IR (a) and ^{13}C NMR spectra (b) of recovered LBGs and GGs.

example, the two main carbon signals near 100 ppm were attributed to the carbon atoms of α-D-galactose and α-D-mannose, respectively. In comparison with original galactomannans (Jia et al., 2024; Wang et al., 2014), the main carbon signals in the recovered galactomannans still existed, and the chemical shift experienced no significant changes. The results of ^{13}C NMR, combined with FT-IR and molecular weight analysis, indicated that the skeletal structure of galactomannan was not destroyed in the recovered LBG and GG. Likewise, Nguyen et al. used the DES system to effectively extract *Cordyceps* polysaccharide, while maintaining its complete skeleton structure throughout the process (Nguyen et al., 2024). Altogether, the dissolution of galactomannans was attributed to breaking and rebuilding hydrogen bond networks in the recovered samples, indicating that galactomannans underwent physical dissolution in the *type II* DESs.

By using SEC-MALLS analysis, the weight average molecular mass (M_W) and polydispersity (PD) of original and recovered galactomannans were determined. As shown in Fig. 6a and Fig. S5, recovered LBG showed no significant changes for M_W (5.442–8.504 × 10^5 g/mol) and PD when compared with original LBG (8.645 × 10^5 g/mol). This indicated that the hydrogen bond interaction played a dominant role in promoting polysaccharide dissolution. For recovered GG, molecular mass data (M_W and PD) were determined as 1.182–2.167 × 10^6 g/mol, which had slightly decreased compared to original samples (2.248 × 10^6 g/mol). This result might be explained by the fact that the higher molecular weight GG chains were not sufficiently dissolved and the lower molecular weight GG chains were collected. Thus, recovered galactomannans still maintained their intact polysaccharide chain structure.

Moreover, monosaccharide compositions and molar ratio analysis were used to further investigate polysaccharide chain structure of recovered galactomannans. As shown in Fig. 6b, two monosaccharide residues constructed the galactomannans chain, including Man and Gal. The molar ratio of M/G for LBG and GG were calculated as 3.63:1 and 1.20:1, respectively. Recovered LBG and GG showed no significant changes for Man/Gal. This result further verified that the chemical stability of recovered galactomannans.

To better understand the dissolution mechanism, molecular dynamics (MD) was used to analyze the conformations, root mean square deviation (RMSD), the number of H-bonds and interaction energies during the dissolution of LBG fragment models in C1E4 (Fig. 7). The ending conformations of 4M-1G models in C1E4 were shown in Fig. 7c. After 0–300 ns, C1E4 destroyed the original hydrogen bond networks of 4M-1G models, and a large number of C1E4 molecules were located between double chains. With the extension of time (300–500 ns), the double chains of 4M-1G models were completely separated and the single chains formed numerous new hydrogen bonds with C1E4. This indicated that the dissolution of LBGs were attributed to the breaking of hydrogen bond networks original galactomannans by C1E4, resulting solubilization. In order to quantitatively characterize the stability of the LBG fragment models in C1E4, we calculated the RMSD values in atomic position of the galactomannan fragment models during simulation as shown in Fig. 7a. Obviously, the curves of 4M-1G models in C1E4 were fluctuated greatly, showing that 4M-1G models were unstable in C1E4 and its atom positions changed significantly during the simulation process. The curves of 4M-1G models also gradually grew (0–300 ns) and stabilized (300–500 ns) with simulation time, which indicated that significant dissolution of galactomannans.

Additionally, hydrogen bonds were a good index to describe the dissolution process of galactomannans, which were divided into intra (4M-1G) and (4M-1G)-C1E4 H-bonds (Fig. 7b). In C1E4, the number of intra (4M-1G) H-bonds decreased rapidly, while the number of (4M-1G)-C1E4 H-bonds increased with simulation time. Meanwhile, little change in H-bonds number was observed in C1E4 after 300 ns, which corresponds to the ending conformations in Fig. 7c. Thus, it indicated that the dissolution of LBG was promoted by breaking intra (4M-1G) H-bonds and rebuilding new (4M-1G)-C1E4 H-bonds. Furthermore, the interaction energies between 4M-1G models and single ions were also calculated, which reflected that the non-bonding energy between the solute and DES components. As shown in Fig. 7d, the interaction of 4M-1G models with Cl^-, Ch^+ and EG was dominated by coulombic interaction, while VDW interaction was quite small. This indicated that the interaction energy (absolute value) of galactomannan fragment models with Cl^- (75.02 kJ/mol) and EG (82.30 kJ/mol) were higher in the C1E4, showing that Cl^- and EG played dominant role in the dissolution process, while Ch^+ (27.84 kJ/mol) played a secondary role. According to the above theoretical calculations, C1E4 featured by multiple hydrogen bond sites which provided the abilities to break the original LBG H-bond networks and rebuild new H-bonds, thereby achieving

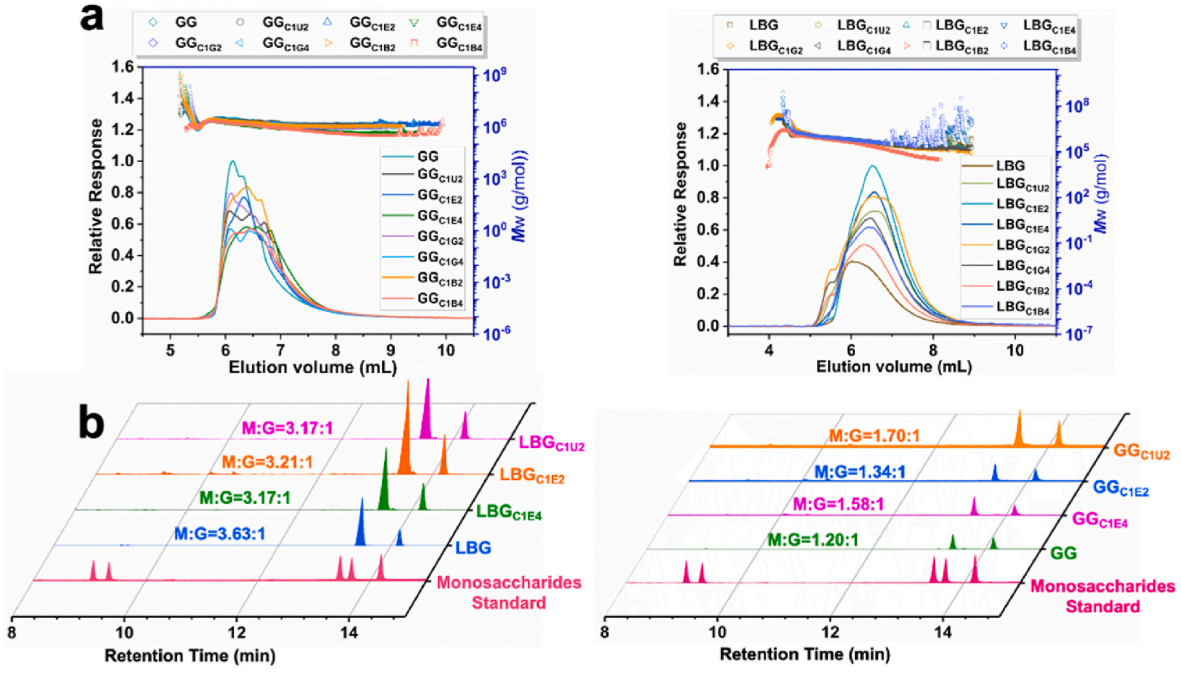


Fig. 6. (a) SEC-MALLS chromatograms and laser light scattering photometry of LBG and GG, (b) Monosaccharide composition analysis of recovered LBG and GG.

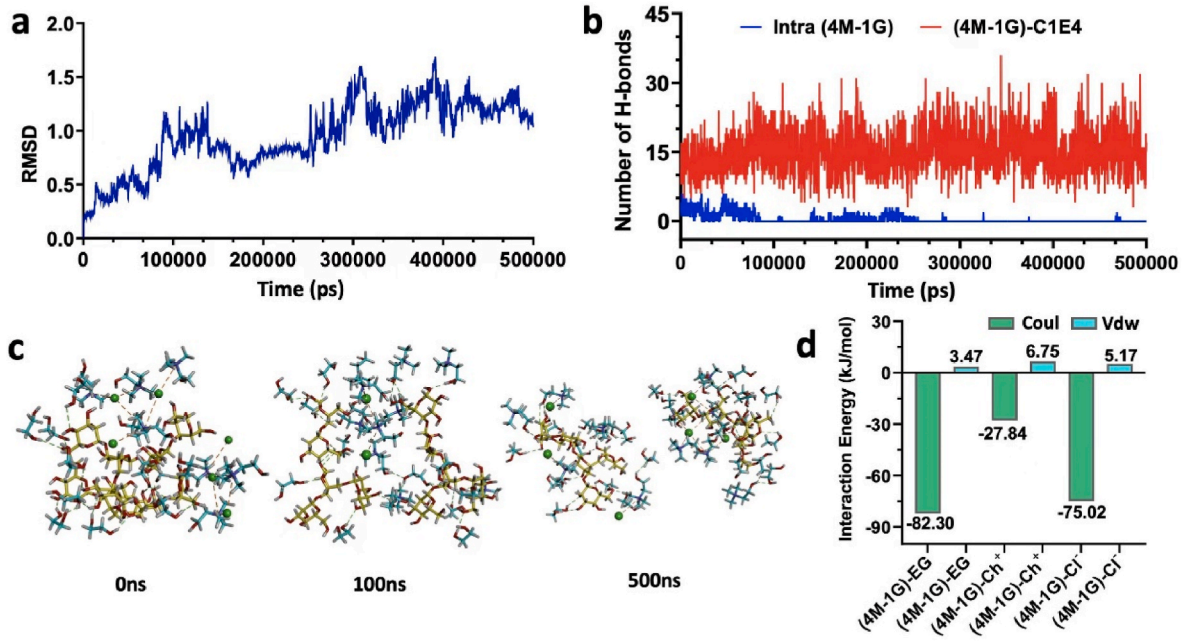


Fig. 7. (a) RMSD of galactomannan fragment models (4M-1G) in C1E4 during simulation, (b) The number of hydrogen bonds in the (4M-1G)-C1E4 system changed with the simulation time: blue color represents intra (4M-1G) H-bonds and red stands for (4M-1G)-C1E4 H-bonds, (c) Configurations of 4M-1G models before and after 500 ns simulation, (d) Interaction energies of different molecular and ions with 4M-1G models in C1E4.

physical dissolution for galactomannans. Similar results were reported by Xia et al., finding that DES effectively disrupted the intricate hydrogen bond network within the lignin-carbohydrate complex, enabling efficient lignin separation (Xia et al., 2018).

In summary, immense anionic and neutral H-bonds supported by *type II* DESs broke and rebuilt galactomannan H-bond networks to exhibit remarkable physical solubilization. Meanwhile, the solubility of galactomannans was related to its molecular weight and distribution pattern of Gal substituents, which could be explained that the repulsive interactions hindered H-bonds breaking due to the high molecular weight and rich Gal side-chains ("hairy" regions) for GG, resulting low solubility (0.16 g/100 mL); compared with the "hairy" regions of GG, LBG had more "smooth" regions (poor Gal side-chains), which promoted high solubility (4.29 g/100 mL) by attractive interaction with DESs (Fig. 8). Similarly, Grisel et al.'s research confirmed that the fine structure of galactomannan significantly affects its interaction with xanthan gum, further validating the crucial influence of different galactomannan structures on its properties (Grisel et al., 2015).

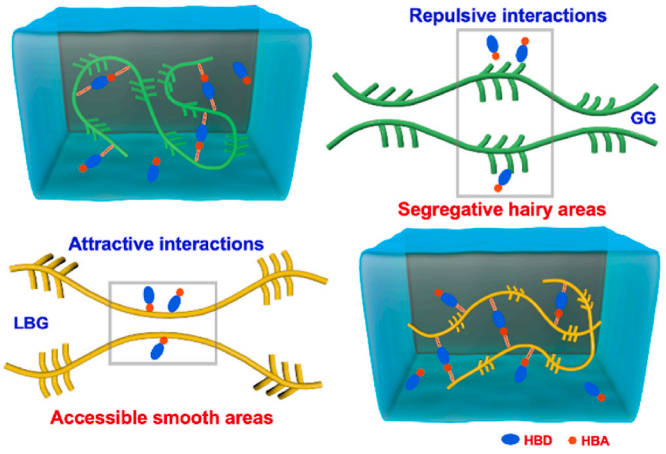


Fig. 8. The mechanism of the structure of galactomannans affecting its solubility.

3.4. Design of type III DESs to promote galactomannans dissolution

To further enhance the dissolution of galactomannans, $AlCl_3{\cdot}6H_2O$ as an anion donor and active acidic site holder was introduced in C1E2 to predict the solubility of *type III* DESs (Fig. 9a). AIM theory showed that ρ values of 4M-2G and 4M-1G models were 0.035 ~ 0.005 au and 0.037 ~ 0.004 au (Table S3), which were smaller than those of C1E2-$AlCl_3{\cdot}6H_2O$ (0.038 ~ 0.005 au). Moreover, $|E_{HB}|$ of C1E2- $AlCl_3{\cdot}6H_2O$ had a large number of weak to medium H-bonds, which were higher than 4M-2G and 4M-1G models. Hence, C1E2- $AlCl_3{\cdot}6H_2O$ broke the H-bonds network of LBG and GG, and reflecting a certain solubility.

Based on the above results of prediction, $AlCl_3{\cdot}6H_2O$ was introduced in selected C1E4 to design *type III* DESs (Fig. 9b). As shown in FT-IR spectroscopy (Fig. 9c), the *type III* DESs emergence of a peak at approximately 1645 cm^{-1} was assigned to the stretching vibration of Al-H, and the signal intensity gradually increased with increasing $AlCl_3{\cdot}6H_2O$ content and appeared a red shift, indicating that the H-bond association in *type III* DESs was enhanced in comparison with the intermolecular H-bond in ChCl and EG. Meanwhile, the force attenuation of intermolecular and intramolecular H-bonds in the C1E4 was illustrated by the decrease in hydroxyl stretching frequency in the region of 3650–3590 cm^{-1}. This implied that the hydroxyls of EG were likely to coordinate with $AlCl_3{\cdot}6H_2O$ because Al three-electron redox properties attracted electron-rich oxygen. The ^1H NMR spectra and ^1H–^1H NOESY further revealed the characteristics of the *type III* DESs. Compared to ChCl, EG and C1E4, the hydrogen atom of H-1 and H-5 shifted to 4.68 ppm (Fig. 9d). Meanwhile, as shown in Fig. 9e, the individual components of ChCl and EG, suggesting intramolecular NOE signals at $\delta_{H/H}$ (4.65/2.97) and $\delta_{H/H}$ (4.65/3.41). Furthermore, it was also found that clear cross-signals at $\delta_{H/H}$ (4.65/4.65) of *type III* DESs. These results illustrated the existence of complex H-bond interaction in *type III* DESs. In addition, as shown in ^{27}Al NMR spectra (Fig. 9f), Al located on $AlCl_3{\cdot}6H_2O$ was found to be high field shifted from 0 to 2 to 0–0.7 ppm, which confirmed that Al^{VI} was bound to EG by the oxygen atom in the form of $[Al(OH)_2(H_2O)_4]^+$ to form $[Al(OH){\cdot}(H_2O)_4{\cdot}(ligand)_n]^+$. Overall, the results indicated that widespread H-bonds formed in *type III* DESs. Likewise, the study by Zhekenov et al. reported that the formation of hydrous *type III* DESs depended on the formation of a large number of

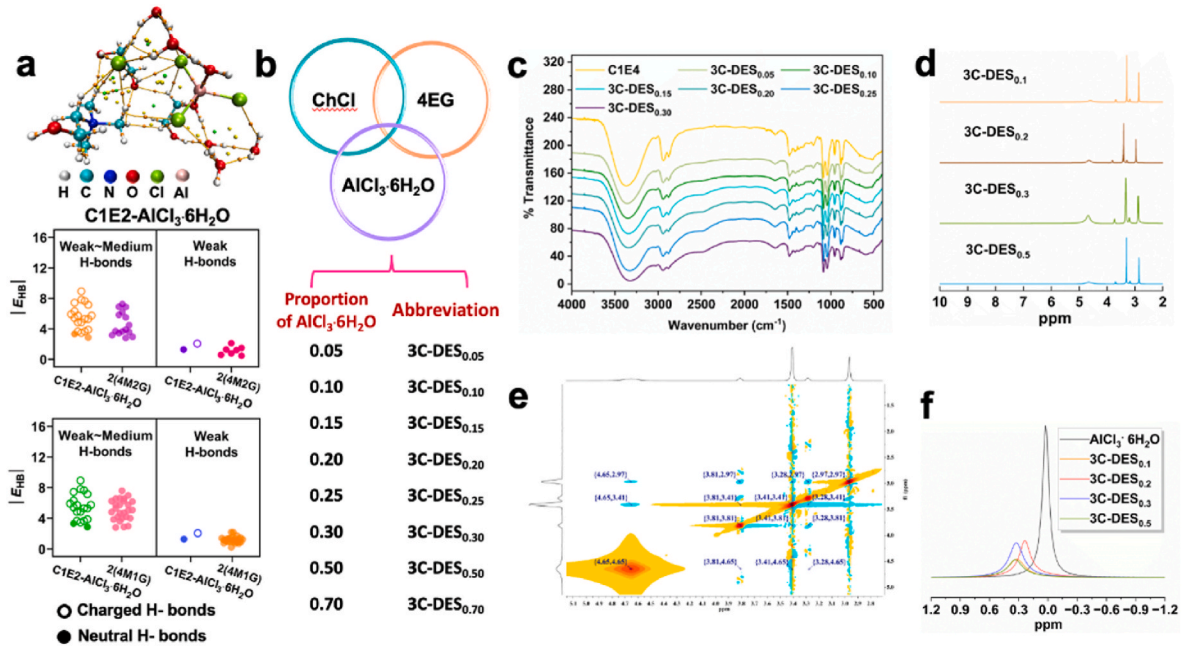


Fig. 9. (a) AIM topological analysis plot of C1E2- $AlCl_3{\cdot}6H_2O$, the absolute value of hydrogen binding energy ($|E_{HB}|$) for GG and LBG fragment models and C1E2-$AlCl_3{\cdot}6H_2O$. The data were calculated at the B3LYP-D3(BJ)/6-311G (d, p) level. All data of AIM were shown in Table S5, (b) The proportion and abbreviation of $AlCl_3{\cdot}6H_2O$ in *type III* DESs, (c) FT-IR of C1E4 and 3C-DESs, (d) ^{1}H NMR spectra of 3C-$DES_{0.1}$, 3C-$DES_{0.2}$, 3C-$DES_{0.3}$, 3C-$DES_{0.4}$ and 3C-$DES_{0.5}$, (e) ^{1}H–^{1}H NOESY spectrum of $DES_{0.2}$, (f) ^{27}Al NMR spectra of $AlCl_3{\cdot}6H_2O$, 3C-$DES_{0.1}$, 3C-$DES_{0.2}$, 3C-$DES_{0.3}$ and 3C-$DES_{0.5}$.

intermolecular hydrogen bonds between the components (Zhekenov, Toksanbayev, Kazakbayeva, Shah, & Mjalli, 2017). Additionally, taking 3C-DES0.1 and 3C-DES0.5 as examples, a viscosity study was conducted to evaluate their performance as well (Fig. S6a). The viscosity was followed order: 3C-$DES_{0.5}$ >3C-$DES_{0.1}$> C1E4 at the same temperature, which demonstrated that the introduction of $AlCl_3{\cdot}6H_2O$ increased the viscosity of *type III* DESs, and the viscosity increases with increasing $AlCl_3{\cdot}6H_2O$ molar ratio.

The above research was found that C1E4 had an excellent performance on the galactomannan solubility, which was significantly influenced by M_w and distribution pattern of Gal substituents. Accordingly, by increasing the amount of $AlCl_3{\cdot}6H_2O$ in C1E4 based on the purpose of constructing multiple hydrogen bond, eight proportional *type III* DESs were formed. The average solubility was increased to 3.91 g/100 mL and 2.54 g/100 mL for LBG and GG compared to C1E4 (Fig. 10a). The dramatic increase in solubility indicated the significant role of $AlCl_3{\cdot}6H_2O$. Moreover, the solubility of LBG and concentration of $AlCl_3{\cdot}6H_2O$ were nonlinearly related and 3C-$DES_{0.15}$ had the greatest solubility. Meanwhile, the solubility of GG increased with the increasing amount of $AlCl_3{\cdot}6H_2O$. This further showed that the amount of $AlCl_3{\cdot}6H_2O$ markedly affected the solubility of galactomannans.

Then, the solution temperature-viscosity curve of LBG and GG dissolved in 3C-$DES_{0.1}$ and 3C-$DES_{0.5}$, respectively, were shown in Figs. S6b–d. The prepared galactomannan/3C-$DES_{0.5}$ solution showed

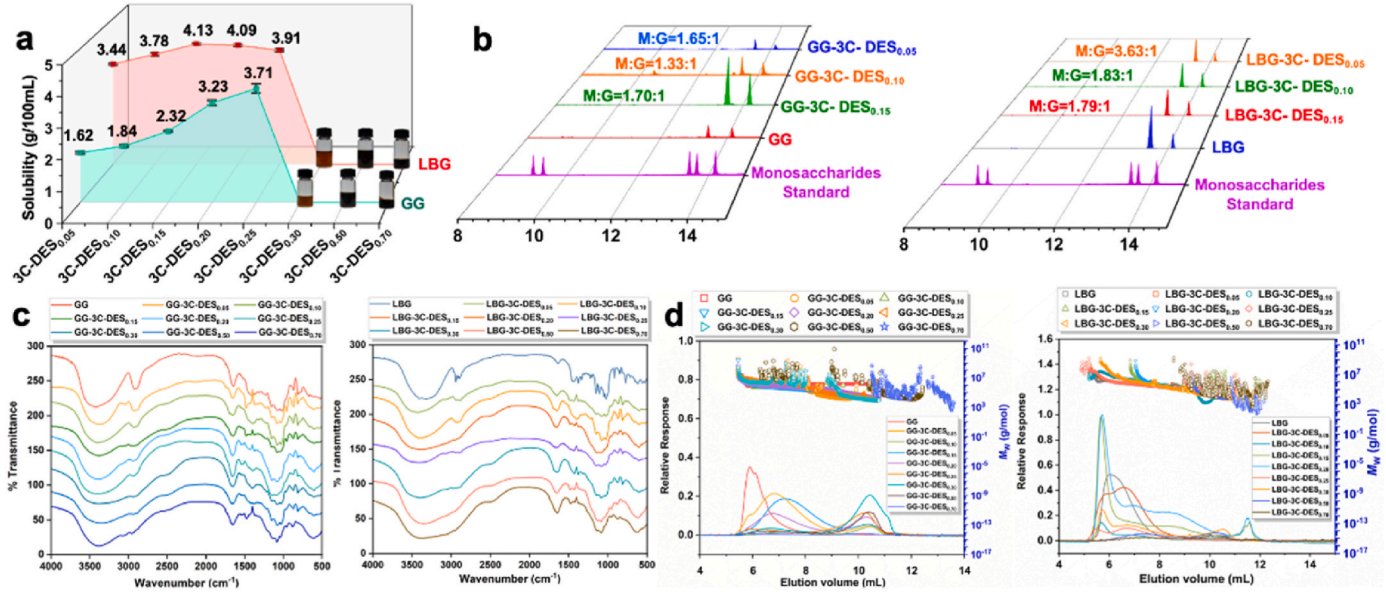


Fig. 10. (a) Solubility of LBG and GG in 3C-DESs, (b) Monosaccharide composition analysis of recovered GG and LBG, (c) FT-IR of recovered LBG and GG, (d) SEC-MALLS chromatograms and laser light scattering photometry of recovered GG and LBG.

higher viscosity than galactomannan/3C-DES$_{0.1}$, and the viscosity decreased with the increase of temperature, which indicated that the viscosity of galactomannan solution with different solvents was mainly affected by the physical and chemical properties of DESs. Moreover, further analysis of different galactomannans viscosity in the same DESs showed that the viscosity of GG/3C-DES$_{0.1}$ was higher than that of LBG/3C-DES$_{0.1}$. Similarly, the viscosity was followed order: GG/3C-DES$_{0.5}$> LBG/3C-DES$_{0.5}$, which further proved that high molecular weight and rich Gal side-chains ("hairy" regions) for GG were not conducive to its flow, thus affecting the viscosity solution.

3.5. *Effect of type III DESs with active acid sites on the structure of recovered galactomannans*

The FT-IR spectra of original and recovered galactomannans were presented in Fig. 10c. It was apparent that three remarkable peaks at around 3421, 1180 and 953 cm^{-1}, which were assigned to the stretching vibrations of O-H, C–O and C–C in galactomannans, respectively. These investigations had shown that the basic skeleton of recovered galactomannans was not destroyed during the dissolution processes. In addition, it was noteworthy that compared to *type II* DESs, the signal intensity of O-H for all recovered samples gradually widened with increasing $AlCl_3{\cdot}6H_2O$ content, which indicated that stronger hydrogen bonding interactions were exhibited between *type III* DESs and galactomannans. Collectively, these results suggested that the hydrogen bond networks of native galactomannans were successfully broken and rebuilt by *type III* DESs, and therefore promoted the solubilization. In a similar study, Ji et al. also demonstrated the effectiveness of DES in disrupting strong intermolecular interactions of polysaccharides. Specifically, they effectively cracked the strong hydrogen bonds and glycosidic bonds in bagasse using *type III* DESs and ultrasonic treatment, resulting in the successful obtainment of nanofibers (Ji, Yu, Yagoub, G. A., Chen, & Zhou, 2021).

Chain characteristics of recovered galactomannans were investigated by molecular mass and monosaccharide composition. As shown in Fig. 10d and Fig. S7, when $AlCl_3{\cdot}6H_2O$ content was increased, recovered LBGs and GGs with different M_W (0.022 ~ 3.776 × 10^5 g/mol and 0.018 ~ 0.177 × 10^6 g/mol) were obtained. Compared to the native galactomannans, all recovered samples showed an obvious decrease of M_W and wide PD values, indicating that the polysaccharide backbone was degraded by a large number of Lewis acid sites from $AlCl_3{\cdot}6H_2O$. These results demonstrated that the M_W of galactomannans was regulated by the molar ratio of $AlCl_3{\cdot}6H_2O$. What's more, monosaccharide compositions and molar ratio analysis further investigated the influence of *type III* DESs on detailed polysaccharide chain structure. As shown in Fig. 10b, Man and Gal still remained the dominant component in recovered galactomannans and the deletion of the main monosaccharide components was not observed. It was worth noting that M/G ratio showed decrease in recovered LBGs and GGs. A possible explanation was that attack of Lewis acid sites from $AlCl_3{\cdot}6H_2O$ on the glycosidic bond promoted the degradation of polysaccharide chains. Taken together, by increasing Lewis acid sites from $AlCl_3{\cdot}6H_2O$, *type III* DESs not only broke and rebuilt the galactomannans hydrogen bond networks, but also interacted with the glycosidic bond and changed the monosaccharide

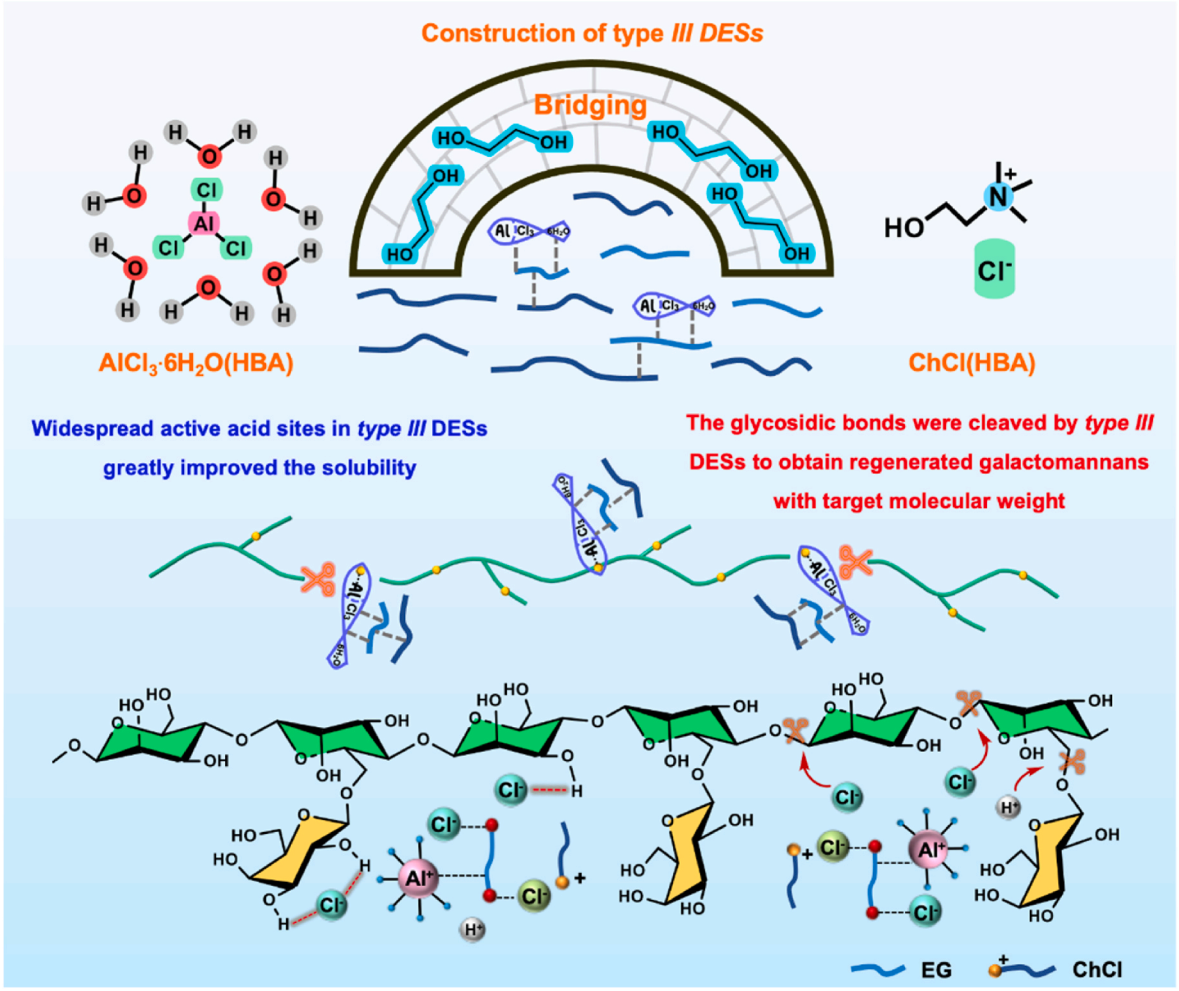


Fig. 11. Schematic diagram elucidating the constitution principle of the 3C-DESs system; the dissolution mechanism of galactomannans in 3C-DESs.

compositions of recovered galactomannans (Fig. 11). With above findings, the *type III* DESs exhibited effective dissolution of galactomannans with different distribution pattern of Gal substituents and molecular weights, while controlling the M_W of galactomannans by adjusting the $AlCl_3{\cdot}6H_2O$ content.

4. Conclusion

In this work, to elucidate how DES components affect the dissolution of polysaccharides with variable molecular weight and Gal substituent distribution, a systematic investigation was conducted using DFT calculation prediction, MD, and characterization of *type II/III* DESs and recovered galactomannans. Among the examined DESs, C1E4 with a high proportion of HBDs exhibited remarkable solubility by providing more H-bond sites. This result explained by the fact that C1E4 with a large number of anionic and neutral H-bonds broke the original galactomannan H-bond networks and rebuilt new hydrogen bonds. The solubility of galactomannans was related to its molecular weight and distribution pattern of Gal substituents, which could be explained that the repulsive interactions hindered H-bonds breaking due to the high molecular weight and rich Gal side-chains ("hairy" regions) for GG, resulting low solubility (0.16 g/100 mL). Additionally, it was indicated that immense number of active acid sites in *type III* DESs not only effectively broke and rebuilt the galactomannans H-bond networks, but also cleaved the glycosidic bond to obtain recovered polysaccharides with target molecular weight. Taken together, this work provided a promising strategy for development of task specific DESs and their widespread polysaccharide resources application in the future.

CRediT authorship contribution statement

Duoduo Guo: Writing – original draft, Data curation. **Jinfeng Han:** Visualization, Investigation. **Hui Tang:** Visualization, Investigation. **Yaxu Sun:** Visualization, Investigation, Data curation. **Li Li:** Visualization, Data curation. **Weibao Kong:** Visualization. **Junyu Liang:** Visualization. **Ji Zhang:** Visualization. **Junlong Wang:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

The present work is supported by the following funds: National Natural Science Foundation of China (No. 51463022), Gansu Education Department (CN) (2020C-21), Gansu Province Scientific Research Fund Project (23JRRA1482, 23JRRA1295), Lanzhou Innovation and Entrepreneurship Project (2023-3-4, 2023-ZD-27), Central Leading Local Science and Technology Development Project (24ZYQA039, 24ZYQA040), Excellent Doctoral Candidates Project of Gansu Province (24JRRA151).

Abbreviation

DES	deep eutectic solvent
3C-DES	the *type III* deep eutectic solvent
GG	guar gum
LBG	locust bean gum
4M-2G	sugar chain fragment models of guar gum
4M-1G	sugar chain fragment models of locust bean gum
M/G	the ratio of galactose to mannose
M_W	weight average molecular mass
SEC-MALLS	size-exclusion chromatographic analysis with Multiangle. laser light scattering detection
ESP	electrostatic surface potential
AIM	atom in molecules
RDG	reduced density gradient
VMD	Visual Molecular Dynamics
HOMO	the highest occupied molecular orbitals
LUMO	the lowest unoccupied molecular orbitals
FT-IR	fourier transform infrared
NMR	Nuclear Magnetic Resonance Spectroscopy
ρ	electron density
$\nabla^2\rho$	laplacian
BCP	bond critical point
PD	polydispersity
$\lvert E_{HB}\rvert$	the absolute values of hydrogen binding energy
MD	molecular dynamics
RMSD	root mean square deviation
C1U2	ChCl-2Urea
C1E2	ChCl-2EG
C1E4	ChCl-4EG
type II DESs	consisting of a quaternary ammonium salt and a metal chloride hydrate
type III DESs	consisting of a quaternary ammonium salt and a HBD

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.foodhyd.2024.110552.

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